



DAMPED SIMPLE HARMONIC MOTION

We know that the motion of a simple pendulum, swinging in air, dies out eventually. Why does it happen? This is because the air drag and the friction at the support oppose the motion of the pendulum and dissipate its energy gradually. The pendulum is said to execute damped oscillations.

In damped oscillations, the energy of the system is dissipated continuously; but, for small damping, the oscillations remain approximately periodic. The dissipating forces are generally the frictional forces. To understand the effect of such external forces on the motion of an oscillator. Let us consider a system as shown in figure below, here a block of mass m connected to an elastic spring of spring constant k oscillates vertically. If the block is pushed down a little and released, its angular frequency of oscillation is ω ($\omega = \sqrt{\frac{k}{m}}$). However, in practice, the

surrounding medium (air) will exert a damping force on the motion of the block and the mechanical energy of the block-spring system will decrease. The energy loss will appear as heat of the surrounding medium (and the block also)

The damping force depends on the nature of the surrounding medium. If the block is immersed in a liquid, the magnitude of damping will be much greater and the dissipation of energy much faster. The damping force is generally proportional to velocity of the bob. [Remember Stokes' Law] and acts opposite to the direction of velocity. If the damping force is denoted by F_d , we have

$$\vec{F}_d = -b\vec{v}$$

where the positive constant b depends on characteristics of the medium (viscosity, for example) and the size and shape of the block, etc. This equation is usually valid only for small velocity. When the mass m is attached to the spring and released, the spring will elongate a little and the mass will settle at some height. This position, shown by O in Figure 1, is the equilibrium position of the mass. If the mass is pulled down or pushed up a little, the restoring force on the block due to the spring is $\vec{F}_s = -k\vec{x}$ where $\vec{x}$ is the displacement of the mass from its equilibrium position. Thus, the total force acting on the mass at any time t , is $\vec{F} = -k\vec{x} - b\vec{v}$. If $a(t)$ is the acceleration of mass at time t , then by Newton's Law of Motion applied along the direction of motion, we have $m a(t) = -kx(t) - b v(t)$

Here we have dropped the vector notation because we are discussing one-dimensional motion. Using the first and second derivatives of $x(t)$ for $v(t)$ and $a(t)$ respectively, we have

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0 \quad \dots(1)$$

Equation 1 is a second-order linear differential equation and its auxiliary equation is $mr^2 + br + k = 0$. The roots are

$$r_1 = \frac{-b + \sqrt{b^2 - 4mk}}{2m} \quad r_2 = \frac{-b - \sqrt{b^2 - 4mk}}{2m} \quad \dots(2)$$

We need to discuss three cases.

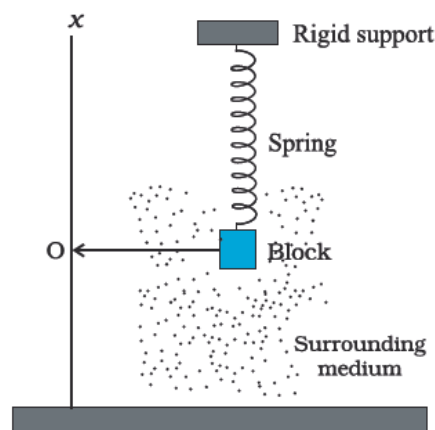
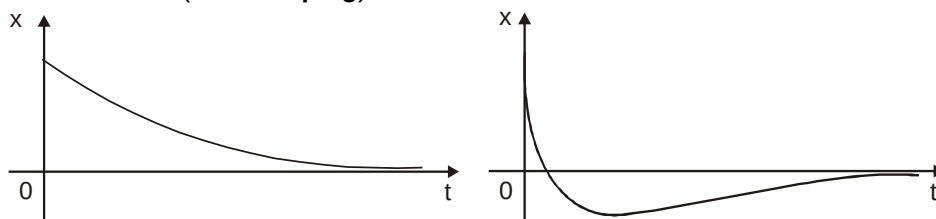


Figure 1: The viscous surrounding medium exerts a damping force on an oscillating spring, eventually bringing it to rest.

**CASE I : $b^2 - 4mk > 0$ (overdamping)**

In this case r_1 and r_2 are distinct real roots and $x = b_1 e^{r_1 t} + b_2 e^{r_2 t}$

Since b , m , and k are all positive, we have $\sqrt{b^2 - 4mk} < b$, so the roots r_1 and r_2 given by Equation 2 must both be negative. This shows that $x \rightarrow 0$ as $t \rightarrow \infty$. Typical graphs of x as a function of t are shown in above figure. Now Notice that oscillations do not occur. (It's possible for the mass to pass through the equilibrium position once but only once.) This is because $b^2 > 4mk$ means that there is a strong damping force (high-viscosity oil or grease compared with a weak spring or small mass).

CASE II : $b^2 - 4mk = 0$ (critical damping)

This case corresponds to equal roots

$$r_1 = r_2 = -\frac{b}{2m}$$

and the solution is given by

$$x = (b_1 + b_2 t) e^{-(b/2m)t}$$

It is similar to Case I, and typical graphs resemble those in above figure, but the damping is just sufficient to suppress vibrations. Any decrease in the viscosity of the fluid leads to the vibrations of the following case.

CASE III : $b^2 - 4mk < 0$ (underdamping)

Here the roots are complex:

$$\left. \begin{matrix} r_1 \\ r_2 \end{matrix} \right\} = -\frac{b}{2m} \pm \omega' i$$

$$\text{Where } \omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \quad \dots(3)$$

The solution is given by

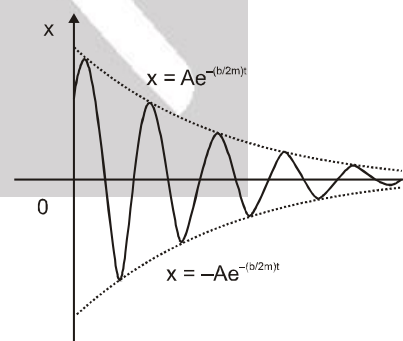
$$x = e^{-(b/2m)t} (b_1 \cos \omega' t + b_2 \sin \omega' t) = A e^{-(b/2m)t} \cos(\omega' t + \phi)$$

We see that there are oscillations the are damped by the factor $e^{-(b/2m)t}$. Since $b > 0$ and $m > 0$, we have $-(b/2m) < 0$ so $e^{-(b/2m)t} \rightarrow 0$ as $t \rightarrow \infty$. This implies that $x \rightarrow 0$ as $t \rightarrow \infty$; that is, the motion decays to 0 as time increases. A graph is shown in figure.

Now the mechanical energy of the undamped oscillator is $\frac{1}{2} k A^2$. For a damped oscillator, the amplitude is not constant but depends on time. For small damping, we may use the same expression but regard the amplitude as $A e^{-bt/2m}$.

$$E(t) = \frac{1}{2} k A^2 e^{-bt/m} \quad \dots(4)$$

Equation shows that the total energy of the system decreases exponentially with time. Note that small damping means that is $\left(\frac{b}{2m}\right)$ much less than ω . Of course, as expected, if we put $b = 0$, all equations of a damped oscillator in this section reduce to the corresponding equations of an undamped oscillator.



**Quality Factor or Q value:**

Practical applications require consideration of the **quality** of the oscillator, Q , which specifies the ratio of total energy, E , to the energy loss, ΔE , over one complete oscillation period, T :

$$Q = 2\pi \frac{E}{|\Delta E|}$$

$$Q = \frac{\omega'}{2\omega_\gamma}$$

where $\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$ and $\omega_\gamma = \frac{b}{2m}$

In the limit of zero damping, the oscillator experiences no energy loss, and $Q \rightarrow \infty$. In the limit of small damping, the quality of the oscillator can be approximated by

$$Q \approx \frac{\omega}{2\omega_\gamma}$$

Combining these two results provides a handy formula for the energy loss during a complete oscillation period of weakly damped motion:

$$|\Delta E| = E \frac{2\pi}{Q} = E \frac{4\pi\omega_\gamma}{\omega} \quad \text{where } \omega = \sqrt{\frac{k}{m}}$$

Example 1. For the damped oscillator shown in Figure the mass m of the block is 200 g, $k = 90 \text{ N m}^{-1}$ and the damping constant b is 40 g s^{-1} . Calculate (a) the period of oscillation, (b) time taken for its amplitude of vibrations to drop to half of its initial value and (c) the time taken for its mechanical energy to drop to half its initial value.

Answer : (a) The time period T from Equation. is given by

$$T = \frac{2\pi}{\omega'} = \frac{2\pi}{\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}} = 0.3 \text{ sec}$$

(b) Now, from equation, the time, $T_{1/2}$ for the amplitude drop half of its initial value is given by

$$A(t) = Ae^{-bt/2m}$$

$$\frac{A}{2} = Ae^{-\frac{bT_{1/2}}{2m}} \Rightarrow T_{1/2} = \frac{2m \ln 2}{b} = 6.93 \text{ s}$$

(c) For calculating the time, $t_{1/2}$, for its mechanical energy to drop to half its initial value we make use of equation from this equation we have.

$$E(t_{1/2}) = \frac{E(0)}{2} \Rightarrow E(0)e^{-bt/m} = \frac{E(0)}{2}$$

$$t_{1/2} = \frac{(\ln 2)m}{b} = \frac{6.93}{2}$$

This is just half of the decay period for amplitude. This is not surprising, because, according to Equation 4, energy depends on the square of the amplitude. Notice that there is a factor of 2 in the exponents of the two exponentials.

Example 2. Show that the system $\frac{d^2x}{dt^2} + \frac{dx}{dt} + 3x = 0$ is underdamped, find its damped angular frequency

Solution :

$$m = 1, b = 1, k = 3$$

$$b^2 < 4mk$$

So, the system is underdamped

$$\omega_d = \sqrt{\left(\frac{k}{m} - \frac{b^2}{4m^2}\right)} = \frac{\sqrt{11}}{2} \text{ s}^{-1}$$





FORCED OSCILLATIONS AND RESONANCE

When a system (such as a simple pendulum or a block attached to a spring) is displaced from its equilibrium position and released, it oscillates with its natural frequency ω , and the oscillations are called free oscillations. All free oscillations eventually die out because of the ever present damping forces. However, an external agency can maintain these oscillations. These are called force or driven oscillations. We consider the

case when the external force is itself periodic, with a frequency ω_d called the driven frequency. A most important fact of forced periodic oscillations is that the system oscillates not with its natural frequency ω , but at the frequency ω_d of the external agency; the free oscillations die out due to damping. A most familiar example of forced oscillation is when a child in a garden swing periodically presses his feet against the ground (or someone else periodically gives the child a push) to maintain the oscillations. Suppose an external force $F(t)$ of amplitude F_0 that varies periodically with time is applied to a damped oscillator. Such a force can be represented as,

$$F(t) = F_0 \cos \omega_d t$$

The motion of a particle under the combined action of a linear restoring force, damping force and a time dependent driving force represented by Equation is given by,

$$m a(t) = -k x(t) - b v(t) + F_0 \cos \omega_d t$$

Substituting d^2x/dt^2 for acceleration in Equation and rearranging it, we get

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = F_0 \cos \omega_d t \quad \dots(5)$$

This is the equation of an oscillator of mass m on which a periodic force of (angular) frequency ω_d is applied. The oscillator initially oscillates with its natural frequency ω . When we apply the external periodic force, the

oscillations with the natural frequency die out, and then the body oscillates with the (angular) frequency of the external periodic force. Its displacement, after the natural oscillations die out, is given by

$$x(t) = A \cos (\omega_d t + \phi) \quad \dots(6)$$

where t is the time measured from the moment when we apply the periodic force. The amplitude A is a function of the forced frequency ω_d and the natural frequency ω . Analysis shows that it is given by

$$A = \frac{F_0}{\sqrt{m^2 (\omega^2 - \omega_d^2)^2 + \omega_d^2 b^2}} \quad \dots(7)$$

$$\text{and } \tan \phi = \frac{-v_0}{\omega_d x_0}$$

where m is the mass of the particle and v_0 and x_0 are the velocity and the displacement of the particle at time $t = 0$, which is the moment when we apply the periodic force. Equation (7) shows that the amplitude of the forced oscillator depends on the (angular) frequency of the driving force. We can see a different behaviour

of the oscillator when ω_d is far from ω and when it is close to ω . We consider these two cases.

- (a) Small Damping, Driving Frequency far from Natural Frequency : In this case, $\omega_d b$ will be much smaller than $m(\omega^2 - \omega_d^2)$, and we can neglect that term. Then Equation. (7) reduces to

$$A = \frac{F_0}{m(\omega^2 - \omega_d^2)} \quad \dots(8)$$

Figure (3) shows the dependence of the displacement amplitude of an oscillator on the angular frequency of the driving force for different amounts of damping present in the system. It may be noted that in all the cases the amplitude is greatest when $\omega_d / \omega = 1$. The curves in this figure show that smaller the damping, the taller and narrower is the resonance peak.

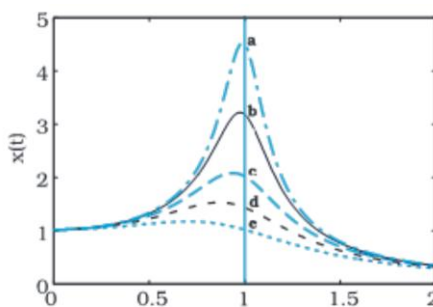


Figure. 3 The graphs illustrate equation (7).
The resonant amplitude ($\omega = \omega_d$) decreases with increasing damping.

If we go on changing the driving frequency, the amplitude tends to infinity when it equals the natural frequency. But this is the ideal case of zero damping, a case which never arises in a real system as the damping is never perfectly zero. You must have experienced in a swing that when the timing of your push exactly matches with the time period of the swing, your swing gets the maximum amplitude. This amplitude is large, but not infinity, because there is always some damping in your swing. This will become clear in the case (b).

- (b) Driving Frequency Close to Natural Frequency : If ω_d is very close to ω , $m(\omega^2 - \omega_d^2)$ would be much less than $\omega_d b$, for any reasonable value of b , then Equation. (7) reduces to

$$A = \frac{F_0}{\omega_d b}$$

This makes it clear that the maximum possible amplitude for a given driving frequency is governed by the driving frequency and the damping, and is never infinity. The phenomenon of increase in amplitude when the driving force is close to the natural frequency of the oscillator is called resonance. In our daily life we encounter phenomena which involve resonance. Your experience with swings is a good example of resonance. You might have realised that the skill in swinging to greater heights lies in the synchronization of the rhythm of pushing against the ground with the natural frequency of the swing.

Example 3. A particle of mass m is attached to a spring (of spring constant k) and has a natural angular frequency ω_0 . An external force $F(t)$ proportional to $\cos \omega t$ ($\omega \neq \omega_0$) is applied to the oscillator. The time displacement of the oscillator will be proportional to :

- (1) $\frac{m}{\omega_0^2 - \omega^2}$ (2) $\frac{1}{m(\omega_0^2 - \omega^2)}$ (3) $\frac{1}{m(\omega_0^2 + \omega^2)}$ (4) $\frac{m}{\omega_0^2 + \omega^2}$

Solution : $x(t) = A \cos(\omega_d t + \phi_0)$

where $A = \frac{F_0}{\sqrt{m^2(\omega^2 - \omega_d^2)^2 + \omega_d^2 b^2}}$

for small damping

$$A \approx \frac{F_0}{[m(\omega^2 - \omega_d^2)]} \Rightarrow x(t) = \left[\frac{F_0}{m(\omega^2 - \omega_d^2)} \right] \cos(\omega_d t + \phi_0) \quad \text{Ans. (2)}$$

Example. 4 In forced oscillation of a particle, the amplitude is maximum for a frequency ω_1 of the force, while the energy is maximum for a frequency ω_2 of the force, then :

- (1) $\omega_1 = \omega_2$ (2) $\omega_1 > \omega_2$
(3) $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large
(4) $\omega_1 < \omega_2$

Solution : Amplitude and energy both are maximum at resonance, when driving frequency is equal to the natural frequency of oscillation. Hence $\omega_1 = \omega_2 = \omega_0$

Ans. (1)



Example 5. If a simple pendulum has significant amplitude (up to a factor of $1/e$ of original) only in the period between $t = 0$ s to $t = \tau$ s, then τ may be called the average life of the pendulum. When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity, with 'b' as the constant of proportionality, the average life time of the pendulum is (assuming damping is small) in seconds :

(1) $\frac{0.693}{b}$

(2) b

(3) $\frac{1}{b}$

(4) $\frac{2}{b}$

Solution: $A(t) = A_0 e^{-bt/2m}$

$$\frac{A_0}{e} = A_0 e^{-bt/2m} \Rightarrow \tau = \frac{2m}{b}$$

In the given question $\frac{b}{m}$ is given as b. $\left(\because a = \frac{-kx}{m} - bv \right)$ **Ans (4)**

Example 6. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to α times its original magnitude, where α equals.

(1) 0.7

(2) 0.81

(3) 0.729

(4) 0.6

Solutions :

$$A(t) = A_0 e^{-bt/2m}$$

Given $A(5) = 0.9 A_0$

$$0.9 A_0 = A_0 e^{-b(5)/2m}$$

$$A(15) = \alpha A_0$$

$$\alpha A_0 = A_0 e^{-b(15)/2m} \Rightarrow$$

$$\alpha = (0.9)^3 = 0.729 \quad \text{Ans (3)}$$

Example 7. Determine whether the system is underdamped, overdamped or critically damped.

(i) $\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 3x = 0$

(ii) $\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = 0$

Solution :

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

If $b^2 < 4mk$ then the system is underdamped

If $b^2 > 4mk$ then the system is overdamped

If $b^2 = 4mk$ then the system is critically damped

(i) overdamped (ii) Critically damped

